Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2019-P1-Model)** **Date: 01-02-2026**

Time: 09.00Am to 12.00Pm **GTA-13** **Max. Marks: 186**

KEY SHEET

PHYSICS

1	**A**	2	**A**	3	**C**	4	**B**	5	**D**	6	**ABCD**
7	**ABCD**	8	**BC**	9	**ABC**	10	**BD**	11	**ACD**	12	**BD**
13	**2**	14	**8**	15	**5**	16	**5**	17	**2**	18	**3**

CHEMISTRY

19	**D**	20	**D**	21	**D**	22	**C**	23	**ABCD**	24	**ACD**
25	**ACD**	26	**CD**	27	**ABCD**	28	**BC**	29	**A**	30	**B**
31	**4.5**	32	**50**	33	**0.25**	34	**100.02**	35	**0.03**	36	**12**

MATHEMATICS

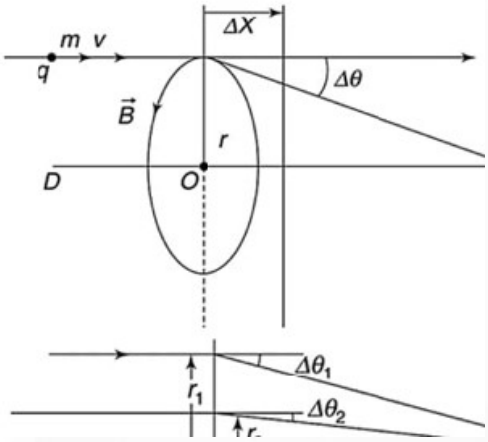
37	**C**	38	**D**	39	**A**	40	**B**	41	**AC**	42	**ACD**
43	**ABCD**	44	**BCD**	45	**BCD**	46	**AC**	47	**AB**	48	**ACD**
49	**7.40**	50	**04**	51	**1**	52	**03**	53	**4975**	54	**638**

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SOLUTIONS

PHYSICS

1. Consider a charge particle at a distance r from the x-axis. Magnetic force on it is towards O (perpendicular) to its original direction of motion. This will cause the path of the charge to deviate by a small angle $\Delta\theta$,



Time required to cross field is $\Delta t = \frac{\Delta x}{V}$ impulse of magnetic force

$= qV(B_0 r)\frac{\Delta x}{V} = qB_0 r\Delta x$ change in momentum of the charge in crossing the field

$\Delta p = qB_0 r\Delta x$ This change is perpendicular to original direction of momentum.

$\Delta p \simeq p\Delta\theta; qB_0 r\Delta r = mV\Delta\theta$

$\Rightarrow \Delta\theta = \frac{qB_0 r\Delta x}{mV}$

Because $\Delta\theta \alpha r$, all ions will get focused at one point F or the axis. (see figure)

$f = \frac{r_1}{\Delta\theta_1} = \frac{r_2}{\Delta\theta_2}; f = \frac{r}{\Delta\theta} = \frac{mV}{qB_0\Delta x}$

2. $\vec{v} = \alpha\left(y\hat{i} + x\hat{j}\right) \qquad \vec{L} = \left(x\hat{i} + y\hat{j}\right)\times\alpha\left(y\hat{i} + x\hat{j}\right) = m\alpha\left[x^2 - y^2\right]\hat{k}$

$\vec{\tau} = \frac{d\vec{L}}{dt} = m\alpha\left(2x\frac{dx}{dt} - 2y\frac{dy}{dt}\right)\hat{k} = 2m\alpha(xv - xv)\hat{k}$

3. Induced emf $\varepsilon = Blv = 2Bxv$

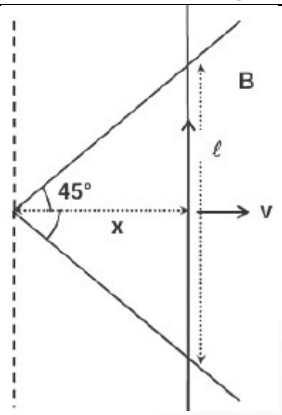
Induced current $I = \frac{\varepsilon}{R} = \frac{2Bxv}{\lambda 2x(\sqrt{2}+1)}$

Magnetic force on the rod $F = -IlB = \frac{-2B^2xv}{2\lambda x(\sqrt{2}+1)}2x$

Acceleration $a = \frac{F}{m} = \frac{-2B^2xv}{\lambda(\sqrt{2}+1)m} \Rightarrow v\frac{dv}{dx} = \frac{-2B^2xv}{\lambda(\sqrt{2}+1)m}$

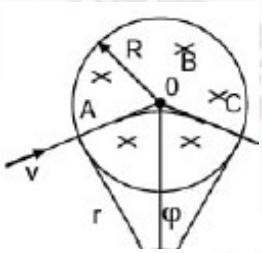
$\Rightarrow \int_{v_0}^{0} dv = -\frac{2B}{m\lambda(\sqrt{2}+1)}\int_0^0 xdx \Rightarrow x_{max} = \frac{\sqrt{m\lambda v_0(\sqrt{2}+1)}}{B}$

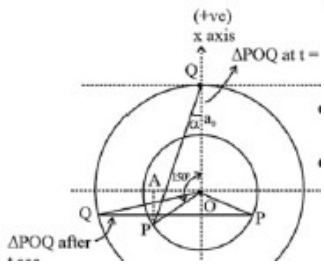
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4. The magnetic induction of the solenoid is directed along its axis, Therefore, the Lorentz force acting on the electron at any instant of time will lie in the plane perpendicular to the solenoid axis. Since the electron velocity at the initial moment is directed at right angles to the solenoid axis, the electron trajectory will lie in the plane perpendicular to the solenoid axis. The Lorentz force can be found from the formula
F = evB.
The trajectory of the electron in the solenoid is an arc of the circle whose radius can be deter mined from the relation $evB = mv^2 / r,$ whence $r = \frac{mv}{eB}$
The trajectory of the electron in shown in figure, where O_1 is the centre of the arc AC described by the electron, v' is the velocity at which the electron leaves the solenoid. The segments OA and OC are tangents to the electron trajectory at points A and C. The angle between v and v' is obviously $\varphi = \angle AO_1C$ since $\angle OAO_1 = \angle OCO_1$.
In order to find φ, let us consider the right triangle OAO_1 : side OA = R and side $AO_1 = r.$ Therefore, $\tan(\varphi / 2) = R / r = eBR / (mv)$.Therefore, $\varphi = 2\tan^{-1}\left(\frac{eBR}{mv}\right)$
Obviously, the magnitude of the velocity remains unchanged over the entire trajectory since the Lorentz force is perpendicular of the velocity remains unchanged over the entire trajectory since the Lorentz force is perpendicular to the velocity at any instant.
Therefore, the transit time of electron in the solenoid can be determined from the relation.

5.





by sin law

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$$\frac{\sin\alpha}{\frac{a_0}{\sqrt{3}}}=\frac{\sin(30-\alpha)}{a_0} \qquad \sqrt{3}\sin\alpha=\frac{1}{2}\cos\alpha-\frac{\sqrt{3}}{2}\sin\alpha$$

$$\frac{3\sqrt{3}}{2}\sin\alpha=\frac{1}{2}\cos\alpha \qquad \tan\alpha=\frac{1}{3\sqrt{3}}$$

6. $E4\pi r^2=\frac{q_0-Q_t}{\varepsilon_0} \quad J=\frac{E}{\rho}=\frac{q_0-Q_t}{4\pi\varepsilon_0\rho r^2}$

$$I=JA=\frac{1}{\in_0\rho}(q_0-Q) \quad I=\frac{dQ}{dt}\Rightarrow\frac{dQ}{q_0-Q}=\frac{dt}{\varepsilon_0\rho}\Rightarrow\frac{Q}{q_0}=1-e^{-t/\varepsilon_0\rho}$$

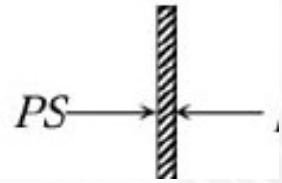
7. Equilibrium of piston gives $PS=kx_0$ or $P=\frac{kx_0}{S}$

Since, the chamber is themally insulated $\Delta Q=0$

$\therefore$ Elastic potential energy of spring = work done by gas

or work done by gas $=\frac{1}{2}kx_0^2$

This work is done in the expensed of internal energy of the gas



8. Conceptual

9. Velocity of particular after collision in vertical direction is 10 m/s $\mu N\Delta t=\mu\times 20m$

For disc, $\mu N\Delta t=mv_c \quad \therefore v_c=4m/s$

$$\mu N\Delta t\frac{R}{2}=\frac{MR^2}{2}(\omega-0) \quad \Rightarrow\omega=20\,rad/s$$

10. Charge flown through S_1 is Q + 3Q = 12 mC

$$0+\frac{(Q+q)}{\varepsilon_0 A}d-\varepsilon\frac{(3Q-q)2d}{\varepsilon_0 A}=0$$

By solving q = 15 mC

11. Conserve angular momentum about O. $2mv_0 2l=1\omega$

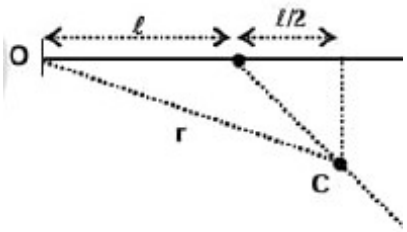
Where $I=\frac{40ml^2}{3} \quad \therefore\omega=\frac{3v_0}{10l}$

$$r=\sqrt{\frac{l^2}{4}+\left(l+\frac{l}{2}\right)^2} \qquad r=\sqrt{\frac{5}{2}}l$$

$$V_c=r_c\omega=\sqrt{\frac{5}{2}}l\times\frac{3V_0}{10l}=\frac{3}{2\sqrt{10}}v_0$$

$$\text{K.E}=\frac{1}{2}I\omega^2=\frac{3}{5}mv_0^2$$

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12. Conceptual

13. For pulse $\frac{dx}{dt}=\sqrt{\frac{T}{\mu}}=\sqrt{\frac{3-10kt}{\mu}}$

$\Rightarrow \int_0^L dx=\int_0^{t_0}\sqrt{\frac{3-10kt}{\mu}}dt$ And $3-10kt_0=0$

14. Extra phase change in glass = phae change in water of length AS_2

$\frac{2\pi t}{\lambda g}-\frac{2\pi}{\lambda\omega}t=\frac{2\pi}{\lambda\omega}.\frac{2}{3}\sin 30^o \quad \Rightarrow t\left[\frac{3}{2}-\frac{4}{3}\right]=\frac{4}{3}\times\frac{1}{3} \quad \Rightarrow t=\frac{8}{3}mm$

15. $2g - T = 2a$ …..(i)

$TR = I\alpha$ ….(ii)

$a = R\alpha$ ….(iii)

From (ii) and (iii) $T=\frac{Ia}{R^2}$

$\therefore\ 2g=a\left(2+\frac{1}{R^2}\right) \quad \Rightarrow\ a=\frac{2g}{\left(2+\frac{I}{R^2}\right)}=\frac{2\times 10}{2+\frac{0.2}{0.01}}=\frac{10}{11}m/s^2$

16. $\sqrt{5gl_2}=\sqrt{gI_1}\Rightarrow\frac{I_1}{I_2}=5$

17. Velocity of efflux $= 16\times 0.25 = 4\,m/s^2$

Time of fall of the liquid $=\sqrt{\frac{2h}{g}}=0.25\sec$

Thus, range on horizontal surface = velocity of efflux x time of fall = 2m.

18. $\Delta T=\left(\frac{\Delta q}{\Delta t}\right)(R)$

Rods connected in series : $\Delta T=\left(\frac{\Delta q}{12}\right)(2R)$ …(1)

Rods connected in parallel : $\Delta T'=\left(\frac{\Delta q}{t}\right)\left(\frac{R}{2}\right)$ …..(2)

$\Delta T'=\frac{1}{4} \quad \therefore\ \Delta T'=\frac{\Delta T}{4}=3\min$

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CHEMISTRY

19. A) KI in oxidized by Cu^{2+} B) Co^{+2} is oxidized by NO_2^-

C) Fe^{+3} is reduced by H_2S D) Fe^{+2} cannot be reduced or oxidized by KCN

20. Al_2O_3 is electrolyically reduced, not through complex formation

21. $\Delta G^o > O \Rightarrow K_{eq} < 1 \because \Delta H > O$

A) $Keq \uparrow with \uparrow in T$ At equilibrium : $[A_2B_3] \neq [A_2B] \because K_q < 1$

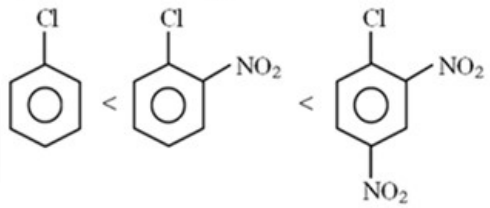
B) $[A_2B_3]$ consumed at T_2 should be more than at T_2 should be more than at T_1 $\because \Delta H > O$

C) [products] can not be more than [reactants] $\because K_{eg} < 1$

D) $[A_2B_3]_{eq}$ can be equal to [B] $1 - x = 2x \Rightarrow K_{eq} = \frac{2}{9}(allowed)$

22.

(A) Rate of ArS_N2



(B) Compounds  can give positive

option C & D reference Peter sykes

23. A) bidantate gives greater CFSE than mono dentate

B) Down the group splitting energy increases (λ decreases)

C) –ve charge on metal makes it a better donar

D) $Pt(NO_2)_2(NH_3)_2$: square planar (6 isomers)

$Zn(NO_2)(NH_3)_2 : Tetra\,hedral$ (3 isomers)

24. A) $H_2SO_4 \rightarrow H_2S_2O_8$ $MnO_4^{2-} \rightarrow MnO_4^-$

$(manganese\,is\,oxidised)$

25. A) $SiCl_4 \xrightarrow{H_2O} H_2SO_3.(d - p\ \pi\ present)$

B) $H_3PO_3 \longrightarrow H_3PO_4 + PH_3\,(no\,new\ \pi\ formed)$

C) $SO_2 \longrightarrow SO_3$

$one(p-d\,bond)$ $(two\ p-d\,bonds)$

D) $Cl_2 \xrightarrow[\Delta]{conc.KOH} Cl^- + \underset{(d-p\,bond)}{ClO_3^-}$

26. A) $C = \sqrt{\frac{3p}{d}}$ with increase in 'p' 'c' does not increase as 'd' also increases proportionally

B) Mean free $path \propto \frac{T}{P}$ (constant if v = constant)

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C) $\overline{KE} \propto T$ D) $C_p = c\ (symmetric\ curve)$

27. A) $(P-Cl)_{axial}$ is longer than $(P-Cl)_{eq}$

B) John teller distortion (d^4)

C) John teller distortion (d^9)

D) 'Cr – O' bond lengths are different in $Cr_2O_7^{2-}$

28. A) 1gm $RaCl_2$ has lesser number of atoms than 1gm of RaO

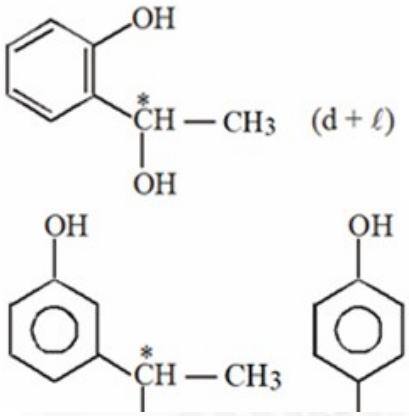
B) 1 mole Ra_3N_2 has more atoms than 1 mol Rao.

C) $U \rightarrow Pb$ (during ase of uranium)

D) $Ra_3(PO_4)_2$: $3^3 3^2 (S)^2 = x$

Donot contain same number of Radium atoms.

29. Since compounds give +ve test with neutral $FeCl_3$ so they must be phenolic



Total 6 isomers

Total 3 racemic mixtures

30. P is Ethanol, Q is 2-Butanol P_1 is Ethyl Bromide P_2 is n-Butane P_3 is 2 Chloro - Butane

31. $P_4 + 3NaOH + 3H_2O \longrightarrow 2NaH_2PO_2 + PH_3 \uparrow$

$3NaH_2PO_2 \xrightarrow{HCl} 3NaCl + 3H_3PO_2$

$3H_3PO_2 \xrightarrow{\Delta} \frac{3}{2}H_3PO_4 + \frac{3}{2}PH_3 \uparrow$

$1mol\ P_4 \Rightarrow \frac{3}{2}H_3PO_4$

$0.1mol\ P_4 \Rightarrow 0.15\,mol\ H_3PO_4$

$0.15 = 0.1 \times V_{lt} \Rightarrow V_{lt} = 1.5$

32. As per equation six molar water are oxidized to '3' moles of O_2

$3O_2 \Rightarrow -12e^- \Rightarrow \left(3xe^{+4} \rightarrow 3xe\right)$

$\therefore$ out of six 'XeF_4' molecular. The remaining '3' undergo disproportionation

$3Xe^{+4} \longrightarrow xe + 2xe^{+6}$

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$A^{+1} \rightarrow A^{2+} + e^-$

33.

$A^+ + e^- \longrightarrow A$

$2A^+ \longrightarrow A + A^{2+}$

$0.5 = \frac{0.06}{1}\log K_{eq}$

$A \longrightarrow A^{2+} + 2e^-$

$A^+ + e^- \longrightarrow A$

$A^+ + e^- \longrightarrow A + A^{2+}$

$E_2^0 = \frac{0.06}{2}\log K_{eq}$

Both cells have same K_{eq} for the reaction.

$\therefore E_2 = \frac{E_1}{2} = 0.25 volts$

34. The reaction is endothermic Heat absorbed in 16 minutes $= 10 \times 30 = 300 cal$

$\therefore$ reaction completed $= \frac{0.3 Kcal}{0.4 Kcal} \times 100 = 75\%$

= two half lives $\Rightarrow t_{1/2} = 8 \min utes$

At the end of 8th minute 50% reaction completed

$Ax_2 \longrightarrow A^{2+} + 2x^-$ $\Rightarrow i = 2$

$1 - 0.5 \quad 0.5 \quad 1$

$\Delta T_b = 0.5 \times (0.1/5) \times 2 = 0.02$

35. $2KMnO_4 + 5H_2O_2 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5O_2 + 8H_2O$

$2H_2O_2 \longrightarrow 2H_2O + O_2$, In the first reaction $5H_2O_2 \Rightarrow 5O_2$

In the decomposition $2H_2O_2 \Rightarrow '1' O_2$

Let 'a' moles H_2O_2 be initially present

$'0.6a' H_2O_2 \Rightarrow '0.6a' mole O_2$. $'0.4a' H_2O_2 \Rightarrow '0.2a' moles O_2$

$\Rightarrow '0.8a' moles 5O_2 = 2.24 lt \Rightarrow 0.1 mol$ $\Rightarrow a = \frac{1}{8} moloes c$

$60\% a = \frac{1}{8} \times 0.6 = \frac{0.6}{8}$ $n_{MnO_4^-} = \frac{0.6}{8} \times \frac{2}{5} = \frac{1.2}{40} = 0.03 mol$

36. Product is $CH_2 = CH - CH = CH_2$

Value of x = 2

Value of y = 2

Value of z= 3

2 x 2 x 3 = 12

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MATHEMATICS

37. Since $f(x_1)=f(x_2)\forall x_1,x_2\in R$ $\therefore$ f(x) must be constant

Hence $p^2-1=0\Rightarrow p=\pm1$ $q^2+2q-3=0$ $\Rightarrow q=-3,1$

$\therefore$ Largest possible value of |p + q| = 4

38. $(1+x)^{101}=\left({}^{101}C_0+{}^{101}C_1x+{}^{101}C_2x^2+.....\right)$

$(1+x)^{100}=\left({}^{100}C_0+{}^{100}C_1x+{}^{100}C_{100}x^{100}\right)$

$\left({}^{100}C_{100}x^{100}+{}^{100}C_{99}x^{99}+.....\right)$ $\therefore$ coefficient of x^{100} is

${}^{201}C_{100}={}^{101}C_0{}^{100}C_{100}+{}^{101}C_1{}^{100}_{99}+.....$

Also $(1-x)^{100}$

$\left({}^{100}C_0-{}^{100}C_1x+{}^{100}C_2x^2....-{}^{100}C_{99}x^{99}+{}^{100}C_{100}x^{100}\right)$

$\therefore(1+x)^{101}(1-x)^{100}=\left({}^{101}C_0+{}^{101}C_1x+{}^{101}C_2x^2.....\right)$

$\left({}^{100}C_{100}x^{100}-{}^{101}C_{99}x^{99}+.....\right)$

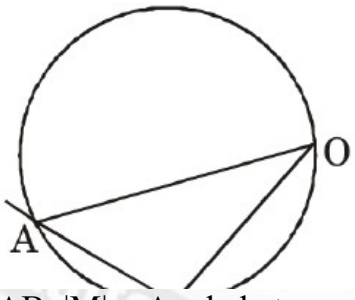
$\therefore coff.of\ x^{100}\ of$

${}^{100}C_{50}={}^{101}C_0{}^{100}C_{100}-{}^{101}C_1{}^{100}C_{99}+{}^{101}C_2{}^{100}C_{98}-.....$

Adding the two

$\Rightarrow{}^{201}C_{100}+{}^{100}C_{50}=2\left[{}^{101}C_0{}^{100}C_{100}+{}^{101}C_2{}^{100}C_{98}+....\right]$

39. Circle touches the x-axis at point B.



Slope of line AB, |M| = Angle between OA & OB

$$|M|=\left|\frac{\frac{1}{2}-\frac{1}{7}}{1+\frac{1}{2}\times\frac{1}{7}}\right|=\frac{1}{3}$$

40. Let $\frac{1}{8}=\int_0^{\pi/2}\frac{x^2(\cos x-\sin x)dx}{(\cos x+\sin x)^3}$ $\Rightarrow\frac{1}{8}=\int_0^{\pi/2}\frac{x^2\cos 2xdx}{(1+\sin 2x)^2}$

Let 2x = t $I=\int_0^{\pi}\frac{t^2\cos tdt}{(1+\sin t)^2}$

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By parts $= \left(\frac{-t^2}{1+\sin t}\right)_0^{\pi} + 2\int_0^{\pi} \frac{t}{1+\sin t}$ $I = -\pi^2 + 2I_1$

Where $I_1 = \int_0^{\pi} \frac{t}{1+\sin t}$ .....(i)

From king property $\Rightarrow \frac{2I}{\pi} = (\tan t - \sec t)_0^{\pi} \Rightarrow \frac{2I_1}{\pi} = 1+1 \Rightarrow I_1 = \pi \Rightarrow I = \pi(2-\pi)$

41. $\frac{e^x + e^{4/x}}{x} > e^x$ for $x \in (1,2)$ and for option (A) put $x = \frac{4}{t}$

42. Lines L_1 and L_2 are intersects at point (1, 0, 2)

43. $P = \frac{(1+2+3+.....10)^2}{2} = \frac{1}{2}\left(\frac{10\times 11}{2}\right)^2 = \frac{55^2}{2}$

44. Let $g(x) = f(x) - 2x^2$

$\Rightarrow g(x) = 0$ has 3 real roots $x = 1, 2$ and 3

$\Rightarrow g'(x) = 0$ has atleast 2 real roots in (1, 3)

$\Rightarrow g''(x) = 0$ has atleast 1 real root in (1, 3)

$\Rightarrow f''(x) - 4 = 0$ has atleast 1 real root in (1, 3)

45. Domain is $x \in [-1,1]$

Given, $\sin^{-1} x = 2\tan^{-1} x$

$\Rightarrow \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) \Rightarrow \frac{2x}{\sqrt{1-x^2}} = \frac{2x}{1-x^2}$

$\therefore x = 0 \, or \left(1-x^2\right)^2 = 4\left(1-x^2\right) \Rightarrow \left(1-x^2\right)\left(3+x^2\right) = 0 \quad \Rightarrow x = -1, 0, 1$

$\tan^{-1}\sqrt{x(x-1)} + \cos ec^{-1}\sqrt{1+x-x^2} = \frac{\pi}{2}$

$x(x-1) \geq 0 \cap x - x^2 \geq 0 \Rightarrow x(x-1) = 0 \Rightarrow x = 0.1$

Now, $(1) \cap (2)$ gives x = 0, 1

Hence, number of common solution are $2\sin^{-1}\sqrt{1-x^2} + \cos^{-1} x = \frac{\sqrt{1-x^2}}{x} - \sin^{-1} x$

Or $\frac{\pi}{2} + \sin^{-1}\sqrt{1-x^2} = \cot^{-1}\frac{\sqrt{1-x^2}}{x}$ $\tan^{-1}\frac{\sqrt{1-x^2}}{x} + \sin^{-1}\sqrt{1-x^2} = 0 \Rightarrow -1 \leq x < 0 \cup \{1\}$

46. Any vector r in space can be written as

$\vec{r} = \lambda\vec{a} + \mu\vec{b} + k(\vec{a}\times\vec{b})$

Take dot product with $\vec{a}\times\vec{b}$. We get $k = \frac{5}{4}$

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$= 2\int_0^{2\lambda} \frac{dx}{x^2+1} = 2\tan^{-1}(2\lambda) = \frac{\pi}{2}$ $\Rightarrow \frac{\pi}{2} = \frac{1}{2}$ and $\mu = -\frac{3}{4}$

47. When $x < \frac{1}{2}$

$(2x-1)^2 - 3(1-2x) + 2 = 0$

$\Rightarrow 4x^2 - 4x + 1 - 3 + 6x + 2 = 0$

$\Rightarrow 4x^2 + 2x = 0$ $\Rightarrow 2x + (2x+1) = 0$

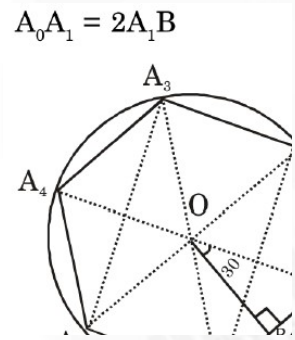
When $x \geq \frac{1}{2}(2x-1)^2 - 3(2x-1) + 2 = 0$

$\Rightarrow 4x^2 - 4x + 1 - 6x + 3 + 2 = 0$ $\therefore x = 1,\ x = \frac{3}{2}$

Hence sum of the roots $= 0 - \frac{1}{2} + 1 + \frac{3}{2} = 2$

48.

$A_0A_1 = 2A_1B$



$= 2(0A_1)\sin\frac{\pi}{6}$ $= 2.1.\frac{1}{2} = 1$

Similarly

$A_0A_2 = 2 \times 1 \times \sin\left(\frac{\pi}{3}\right) = \sqrt{3}$

And $A_0A_4 = A_0 \times A_2 = \sqrt{3}$

Thus $A_0A_1 \times A_0 \times A_2 \times A_0A_4 = 3$

Area of regular hexagon $6\times$ Area of ΔA_0OA_1

$= 6 \times \frac{1}{2} \times A_0A_1 \times OB$ $= 6 \times \frac{1}{2} \times 1 \times (OA_1)\cos\frac{\pi}{6}$

$= 6 \times \frac{1}{2} \times 1 \times 1 \times \frac{\sqrt{3}}{2}$ $= \frac{3\sqrt{3}}{2}$

Area of quadrilateral $A_0A_2A_3A_5 = (A_0A_2)$

(A_2A_3) $= \sqrt{3} \times 1$ $= \sqrt{3}$

49. $f'(0^-) = 1$ & $\lim_{x\to 0} \frac{1 - \cos\left(1 - \cos\frac{x}{2}\right)}{2^m x^n} = 1$

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$$\lim_{x\to 0}\frac{\left(1-\cos\frac{x}{2}\right)^2}{2^m x^n}=2 \qquad \therefore \text{ for limit to be exist n = 4}$$

$$\Rightarrow \frac{2}{2^{8+m}}=1\Rightarrow m=-7 \text{ and} \qquad \text{Hence (n – 10 m) = 74}$$

50. $z_1=r_1^{i\theta 1}\ and\ z_2=r_2e^{i\theta 2}\Rightarrow \frac{z_1\bar{z}_2+\bar{z}_1z_2+z_1z_2+\bar{z}_1\bar{z}_2}{|z_1z_2|}=2\cos(\theta_1-\theta_2)+2\cos(\theta_1+\theta_2)$

51. Given a + b + c = 1 ……1)
9a + 3b + c = 7 ……..2)
18 < 25a + 5b + c < 22 ……3)
$\Rightarrow$ From above (1), (2) and (3) 4 < 7a – b – c < 8
4 < 7a – b + a + b – 1 < 8
5 < 8a < 9
$\frac{5}{8}<a<\frac{9}{8}(a=1)$
A = 1, b = -1, c = 1
For question (2) h(x) $=\ln(x^2-x+1)-x$

52. $L=\frac{2}{3}\times e^2$

53. $g(x)=[x]+\sqrt{\{x\}}$
$g'(x)=\frac{1}{2\sqrt{x}}$
x = ¼
$sum=\sum_{r=0}^{99} r+\frac{1}{4}$

54. $\left(2+\frac{x^4}{x^2+1}\right)^{10}=\left(\frac{x^4+2x^2+2}{x^2+1}\right)=\left[\left(x^2+1\right)+\frac{1}{x^2+1}\right]^{10}$

$\therefore\left(\left(x^2+1\right)+\frac{1}{\left(x^2+1\right)}\right)^{10}={}^{10}C_0\left(x^2+1\right)^{10}+{}^{10}C_1\left(x^2+1\right)^{8}+{}^{10}C_2$

$\left(x^2+1\right)^6+.....{}^{10}C_5+\frac{{}^{10}C_6}{\left(x^2+1\right)^2}+.....\frac{{}^{10}C_{10}}{\left(x^2+1\right)^{10}}$

$\therefore A_0={}^{10}C_0+{}^{10}C_1+{}^{10}C_2+{}^{10}C_3+{}^{10}C_4+{}^{10}C_5=638$

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